

# CHGME: A Cross-Modal Sentiment Analysis Model Targeting Online Public Opinion in Higher Education Institutions

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## Abstract

With the widespread adoption of the internet and social media, the analysis of online public opinion in universities has become a critical issue in social governance and educational management. The diversity and complexity of public opinion data pose significant challenges for sentiment analysis: (1) Existing publicly available multimodal datasets primarily focus on daily-life scenarios, lacking topic relevance to university contexts and targeted corpora. (2) The effective integration of multimodal information (e.g., audio, text, and video) to accurately capture emotional dynamics remains understudied. (3) The ability to leverage synergistic effects while avoiding interference across different tasks and modalities is often overlooked. To address these challenges, this paper proposes a multimodal sentiment analysis model for university-related online public opinion, leveraging a Mixture of Experts (MoE) framework to fuse the strengths of multimodal data. Experimental results demonstrate that CHGME outperforms baseline models on both the CMU-MOSEI dataset (by 0.3%) and a self-constructed Public Safety Dataset (PSD) (by 1.97%) in multimodal sentiment analysis tasks.

## CCS Concepts

• **Internet** → Web services; Network monitoring.

## Keywords

multimodal, mixture of experts (MoE), online public opinion, sentiment analysis

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## 1 INTRODUCTION

Social media and short video platforms have become primary channels for university stakeholders to express opinions and emotions. The rapid spread of negative or misleading content threatens campus stability, while traditional text-based sentiment analysis is often inadequate for the sensitive and emotionally direct nature of campus-related discourse. Multimodal data—text, audio, and video—carries richer emotional cues and has grown predominant. Developing tailored multimodal sentiment analysis for universities can therefore enhance public-opinion monitoring and support intelligent campus governance.

To tackle key challenges such as data scarcity, modal heterogeneity, and insufficient fusion mechanisms, this paper proposes CHGME (Cross-modal Hierarchical Gated Mixture of Experts), a model designed for university public safety. CHGME introduces a video-guided dynamic gating strategy that strengthens text relevance while suppressing audio-visual noise. The main contributions are: (1) a self-built multimodal dataset (PSD) containing 1,000 university-related samples; (2) a Hierarchical Gated Mixture of Experts (HGME) for flexible multimodal fusion; and (3) a content-aware fusion mechanism that adjusts dynamically based on video confidence scores.

## 2 RELATED WORK

Public opinion now integrates text, audio, images, and video, necessitating multimodal approaches to fully capture complex emotional expressions. This shift from single-modality analysis has led to the adoption of structures like Mixture of Experts (MoE) to enhance modeling of heterogeneous data. In judicial public opinion, Guo et al. [1] improved Weibo comment classification by embedding emotional knowledge graphs, while Zhao et al. [2] combined BERT with hierarchical attention for better emotion perception in complex contexts. For emergencies, Han et al. [3] developed a multi-channel feature fusion model to identify extreme emotions in public health-related Weibo texts. These methods face generalization challenges in real-world multimodal settings.

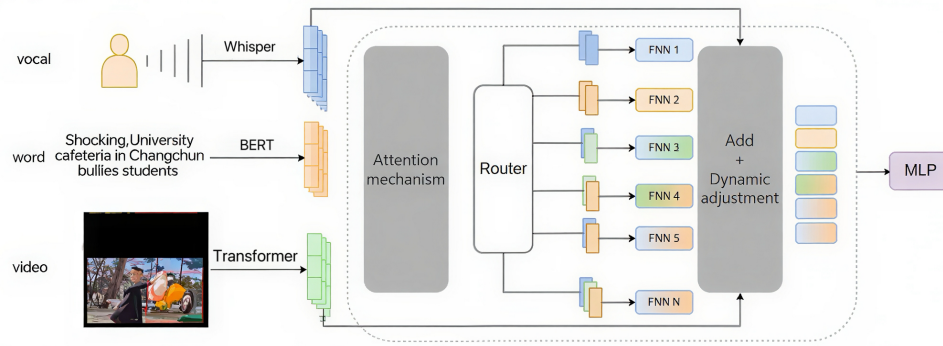


Figure 1: Model Framework Diagram

Multimodal approaches have advanced with models like Lu et al. [4], which integrated audio, image, and text via attention mechanisms for fine-grained emotion modeling. Han et al. [5] incorporated BERT’s multilayer features to capture non-textual cues in short video comments. Song et al. [6] further proposed a multi-level hybrid fusion framework for joint emotion transfer learning in multitask scenarios. Gandhi A et al. [7] presents a systematic review of multimodal sentiment analysis. Ghosal et al. [8] introduced context-aware cross-modal attention, and Zadeh et al. [9] designed multi-attention recurrent networks to model temporal dependencies. Akhtar et al. [10] extended this with a multi-task contextual attention framework for shared emotion-related learning. For emotion intensity, Yang et al. [11] proposed a contrastive learning framework (CLGSI) to enhance extreme emotion recognition. You et al. [12] fused visual and textual data for stronger semantic interaction. Yu and Jiang [13] improved target-specific emotion judgment via object-oriented multimodal BERT.

In structural modeling, Zadeh et al. [14] introduced tensor fusion networks and Tsai et al. [15] designed multimodal Transformers to address temporal alignment. These methods have shown strong performance on benchmarks like CMU-MOSI and CMU-MOSEI. MoE mechanisms have gained prominence for handling multimodal heterogeneity. Chi et al. [16] demonstrated that sparse-activated MoE reduces representation collapse and improves feature diversity. Chowdhury et al. [17] optimized computational efficiency with patch-level routing. Huddar et al. [18] propose an attention-based multimodal fusion framework and Zhang et al. [19] introduce a novel multimodal benchmark. Zhang et al. [20] enhanced training stability via gradient-adjusted robust MoE frameworks. In language modeling, Dai et al. [21] developed DeepSeekMoE for parameter-efficient text generation, while Li et al. [22] addressed label noise with self-supervised single-modality labeling. Zhang et al. [23] improved cross-modal alignment via dual-branch contrastive learning, and FuseMoE [24] enabled flexible cross-modal expert routing. Huang et al. [25] further introduced task-difficulty-aware routing to boost model performance. Pan et al. [26] survey deep multimodal representation learning, covering fusion/alignment techniques and future directions in cross-modal feature extraction. Yu et al. [27] propose a two-stage attention-based fusion method for image-text sentiment contrastive learning, enhancing cross-modal interaction and analysis robustness. Feng et al. [28] develop a top-down

mask-guided cascading Transformer framework for multimodal sentiment analysis, improving feature integration and classification accuracy.

### 3 SYSTEM ARCHITECTURE

This section details the implementation of the CHGME (Cross-modal Hierarchical Gated Mixture of Experts) model for multimodal sentiment classification in campus safety (see Figure 1). CHGME employs a hierarchical gating mechanism that routes audio, text, and video through dedicated expert networks to filter noise and reduce computation. It enriches emotional representations by integrating fine-grained cues such as facial expressions and vocal intonation. Finally, a dynamic gating mechanism selects and weights the outputs of modality-specific experts, fusing them via weighted summation to generate the final sentiment classification.

Assume  $K$  is a set of multimodal data, where each data instance  $u_i \in K$  consists of a sentence  $X_i \in \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ , along with its corresponding audio  $A_i$  and video  $M_i$ . Here,  $n$  denotes the number of words. The task of multimodal sentiment analysis for campus public safety is to predict the sentiment polarity label  $y_i \in \{-2, -1, 0, 1, 2\}$  for the video  $M_i$ . The objective of this paper is to learn a function  $f: (X_i, A_i, M_i) \rightarrow y_i$ . For example, given a video such as “Shocking: A university canteen contractor in Changchun bullies students,” the model should be able to predict a sentiment tendency of -2.

During the audio feature extraction stage, the system employs the Whisper-Large-v2 model to extract phonemic and prosodic features from the audio. First, the audio signal  $A_i$  is converted into a Mel spectrogram  $\Lambda_i$ . Then, a hybrid model combining convolutional neural networks (CNN) and Transformer architectures is utilized to capture the phonemic features  $\tau_i$  and prosodic features  $\sigma_i$  from the audio. Ultimately, the Whisper-Large-v2 model outputs a 1024-dimensional acoustic fingerprint vector  $A'$ , which represents the emotional information in the audio, including emotional tone, intonation, speech rate, and other characteristics.

In the text feature extraction stage, the BERT-Base model was selected as the pre-trained text model  $X_{pdue}$  due to its ability to deeply encode input text, capturing rich semantic and emotional information as well as contextual dependencies. After processing, the text data is transformed into a 768-dimensional dynamic contextual vector  $X'$ .

In the video feature extraction stage, a Transformer model is employed to process the video data. The video is first converted into image frames  $M^p$ , and features are extracted from each frame using the Transformer model. Ultimately, the model outputs a 1024-dimensional video feature vector  $M'$ , which encompasses emotional information from the video, such as changes in facial expressions and emotional cues from body movements. CLIP is used as a candidate model in this process.

Given the central role of video, CHGME introduces a video-gated dynamic weighting mechanism. Unlike standard cross-modal transformers, it adaptively strengthens textual influence while attenuating noise from audio and video based on real-time prediction confidence.

The gating mechanism dynamically adjusts based on video prediction confidence. When uncertainty is high (flat probability distribution), the gate opens to integrate complementary audio and text cues. Conversely, when video confidence is strong, the gate closes to suppress less relevant modalities and minimize interference.

$$u_i = x_i^T x_i \quad (1)$$

Among them,  $x_i$  represents the video-text joint feature corresponding to the current token, and  $u_i \in (0, 1)$  denotes the video gating value for the  $i$ -th token. Based on the multimodal gate, the fused representation for the  $i$ -th token is obtained as follows:

$$\tilde{u}_i = u_i t_i + (1 - u_i) \tilde{x}_i \quad (2)$$

Among them, for the  $i$ -th token,  $u_i$  represents the video gate,  $1 - u_i$  denotes the multimodal fusion confidence gate, and  $t_i$  indicates the time.

$$u_i = \text{Sigmoid} \left( W_g^T [F_{\text{align}} : H_{\text{hist}}] \right) \quad (3)$$

Here, Sigmoid serves as the activation function, outputting gating values between 0 and 1.  $W_g$  is a learnable weight matrix, and  $[F_{\text{align}} : H_{\text{hist}}]$  denotes vector concatenation.  $H_{\text{hist}}$  is the historical modal interaction memory unit, which caches cross-modal information from previous tokens. Through a gated cyclic update strategy, it adaptively adjusts weight sparsity to suppress noise interference from secondary modalities.

First, the preprocessed text, audio, and visual features are captured and subjected to cross-modal concatenation. Subsequently, linear projection is employed to align the concatenated features into a unified latent space, forming a cross-modal interaction base vector:

$$K_{\text{align}} = \text{LayerNorm} \left( K_t \oplus K_a \oplus K_v + \text{CrossAttention}(X, A, M) \right) \quad (4)$$

Here,  $K_{\text{align}}$  represents the fused output (before projection),  $K_t$  denotes the BERT text features,  $K_a$  refers to the Whisper audio features,  $K_v$  stands for the Transformer video features, and CrossAttention indicates the 8-head cross-attention module.

$$F_{\text{final}} = \sum_{i=1}^3 g_i \cdot E_i (F_{\text{align}}) + \text{LayerNorm}(\text{MHCA}(F_{\text{align}})) \quad (5)$$

Here,  $F_{\text{final}}$  represents the final sentiment classification feature,  $g_i$  denotes the normalized weights output by the gating network with  $\sum g_i = 1$ ,  $E_i$  refers to the  $i$ -th feedforward expert network,  $F_{\text{align}}$  is the unified latent vector after linear projection (output of

**Table 1: Number of Samples in the CMU-MOSEI Dataset**

| Model    | CMU-MOSEI |       |      |
|----------|-----------|-------|------|
|          | train     | valid | test |
| Positive | 8048      | 932   | 2284 |
| Neutral  | 3540      | 433   | 1035 |
| Negative | 4738      | 506   | 1350 |

**Table 2: Number of Samples in the PSD Dataset**

| Model    | PSD   |     |      |
|----------|-------|-----|------|
|          | Train | Val | Test |
| Positive | 352   | 41  | 100  |
| Neutral  | 155   | 19  | 45   |
| Negative | 207   | 22  | 59   |

Equation 4 reprojected), and MHCA stands for the 8-head multi-head cross-attention. The outputs of the experts are weighted by the gating mechanism, and the multi-head cross-attention mechanism is then employed to achieve global information integration.

## 4 EXPERIMENTAL RESULTS AND ANALYSIS

### 4.1 Dataset construction

The CMU-MOSEI dataset is adopted for direct performance comparison with existing methods. Additionally, we introduce a self-constructed PSD dataset, whose annotation was independently performed by three annotators specializing in public opinion analysis. Following a predefined labeling guideline, they achieved substantial inter-annotator agreement (Cohen’s  $\kappa = 0.82$ ). Divergent cases were resolved through group consensus. Dataset statistics are shown in Table 1.

A multimodal campus safety dataset—PSD (Public Safety Dataset)—with approximately 1,000 synchronized text, audio, and video samples was constructed to support precise sentiment analysis. Each modality is carefully annotated to enable complementary cross-modal learning, providing a foundation for validating the CHGME model. Dataset statistics are shown in Table 2.

### 4.2 Comparative Test

This subsection compares our proposed model with several representative baselines. Following established practice, we adopt Acc-5, F1, and MAE as evaluation metrics. A prediction is counted as correct only when both the video and its sentiment polarity are correctly matched. The results are summarized in Table 3.

CLGSI is adopted as the baseline model for multimodal processing, as it effectively captures and fuses emotional information through emotion intensity-guided contrastive learning. For comprehensive comparison, recent multimodal methods are also included as baselines.

As shown in Table 3, while CHGME exhibits slight performance variations on the general CMU-MOSEI dataset—partly due to its contextual mismatch with campus safety scenarios—it significantly outperforms all baselines on the domain-specific PSD dataset across

**Table 3: Comparison of performance of PSD and CMU-MOSEI across different models**

| Model        | PSD CMU-MOSEI |       |       |       |       |      |
|--------------|---------------|-------|-------|-------|-------|------|
|              | Acc-5         | F1    | MAE   | Acc-5 | F1    | MAE  |
| TFN [20]     | 51.3          | 65.32 | 49.3  | 40.9  | 65.32 | 54.6 |
| MuIT [21]    | 52.98         | 63.26 | 47.98 | 41.8  | 62.26 | 57.5 |
| Self-MM [22] | 50.53         | 66.37 | 50.53 | 41.3  | 66.37 | 56.9 |
| ConFEDE [23] | 56.3          | 65.78 | 56.3  | 40.2  | 65.78 | 57.4 |
| CLGSI [11]   | 57.42         | 66.38 | 58.42 | 42.6  | 66.38 | 57.2 |
| CHGME        | 59.39         | 66.86 | 63.39 | 42.9  | 66.86 | 58.1 |

**Table 4: Ablation Experiments of CHGME**

| Model | PSD   |       |       |      |
|-------|-------|-------|-------|------|
|       | Acc-5 | F1    | MAE   | Corr |
| -CHG  | 49.3  | 65.32 | 49.3  | 40.9 |
| -ME   | 47.98 | 63.26 | 47.98 | 41.3 |
| CHGME | 50.53 | 66.37 | 50.53 | 41.8 |

key metrics such as Acc-5 and F1. These results validate the effectiveness of CHGME in multimodal sentiment analysis and highlight its robustness in heterogeneous settings, attributable to its hierarchical gated mixture-of-experts mechanism, which enables fine-grained and dynamic modality fusion.

### 4.3 Ablation experiment

To further validate the effectiveness of individual components in the CHGME model, this paper designs ablation experiments by gradually removing or replacing key modules, with the results presented in the table. By comparing classification accuracy (Acc-5), F1 scores, as well as mean absolute error (MAE) and correlation coefficient (Corr) for regression tasks under different configurations, the study systematically evaluates the impact of each component on overall performance. This provides a comprehensive analysis of the model’s robustness in cross-task scenarios, as detailed in Table 4.

As shown in Table 4, removing any module leads to a decline across all evaluation metrics, confirming the effectiveness of the proposed model. Specifically, discarding the video relevance module reduces performance, underscoring the need for video noise control. Removing the Mixture of Experts (MoE) component also lowers results, highlighting how its flexible expert selection and weighting improve both the accuracy and robustness of multimodal fusion.

### 4.4 Experiment Discussion

CHGME advances multimodal design in three aspects: granularity, gating, and noise suppression (Table 4). Operating at the token level, it routes each word through eight parallel experts based on video-text joint features, enabling finer emotion modeling compared to modality-level (FuseMoE) or sample-level (CLGSI) approaches.

Its video-guided dynamic gating adjusts multimodal fusion in real time based on video confidence: lower confidence opens the

gate to text/audio cues, effectively turning uncertainty into an adaptive fusion switch. A Hadamard-gated memory further filters out noise identified by the video gate, following a “select-then-filter” denoising strategy.

On the PSD dataset, CHGME outperforms FuseMoE and CLGSI by 6.09% and 1.97% in Acc-5, respectively. Ablation studies confirm that the video gating mechanism contributes most to this performance gain, underscoring its practical value.

Nevertheless, the limited scale of PSD (1,000 samples) may constrain model generalizability and raise overfitting risks. Although internal annotation consistency was ensured (Cohen’s  $\kappa = 0.82$ ), the lack of an external validation set limits robustness assessment across scenarios. Future work will expand the dataset with more diverse samples and establish a cross-domain validation set to improve real-world applicability.

## 5 CONCLUSION

This paper presents a multimodal sentiment classification model designed for campus public safety applications. It introduces a self-constructed, aligned multimodal dataset (PSD) of approximately 1,000 audio-text-video samples. The proposed model incorporates a video-gated dynamic fusion mechanism that adjusts multimodal integration based on video prediction uncertainty, enabling finer feature control and better cross-modal alignment, which yields a more robust and discriminative representation for sentiment classification.

While CHGME performs well on benchmark datasets, its robustness under video quality degradation requires further validation. Future work will focus on expanding the dataset scale, refining the annotation protocol with external validation, and developing adaptive fusion strategies for low-quality multimodal inputs to improve generalization in real-world environments.

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